

Advanced Python Programming: Final Project Report

Twitter Dataset Analysis, Visualization and Insights extraction (2018-2022)
During the Mati Wildfire

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1 Introduction

This report details the computational analysis of Twitter data related to the 2018 Mati Wild-fire in Greece. Using Advanced Python techniques, we analyzed a dataset comprising author information, timestamps, Greek text, and engagement metrics to understand information dissemination and user behavior during the catastrophe.

2 Project Description

The project objective is to apply data science methodologies to identify patterns of communication, detect temporal bursts of activity, and analyze author influence or suspicious manipulative activity.

- **Text Preprocessing:** Cleaning, normalization, and lemmatization of Greek tweets.
- **Filtering:** Distinguishing relevant content from noise and irrelevant tweets using keyword matching.
- **Activity Analysis:** Measuring tweet frequency, temporal patterns.
- **Influence Detection:** Identifying key drivers of activity bursts, bots and other suspicious activity for political leverage by certain people/groups along with coordinated activity.

3 Methodology

3.1 Filtering and Preprocessing (Task 1)

The raw Twitter dataset includes various noise, such as songs, different wildfires or unrelated locations that mention similar keywords. Therefore, a preprocessing and filtering pipeline was required to isolate tweets that are truly related to the Mati wildfire.

Text preprocessing was applied to remove URLs, mentions, emojis, punctuation, and accents, while normalizing text to lowercase. Greek language tweets were retained, and lemmatization (el_core_news_sm) was applied to normalize different grammatical forms of words.

A tweet was classified as relevant only if it contained: - at least one location-related keyword (e.g. mati, rafina) AND - at least one event, authority, or casualty-related keyword (e.g. fire, evacuation, deaths, coast guard). Note that these words are all greek keywords.

This logical rule-based filtering prioritizes precision over recall, ensuring that only highly relevant tweets are passed to subsequent questions. Around 27% were classified as relevant, forming the base dataset used in all following questions.

3.2 Temporal Burst Detection (Task 2)

To analyze the chronological evolution of the dataset, we calculated the daily tweet volume after filling 534 missing dates to ensure time-series continuity. The methodology utilized a 7-day rolling average to smooth short-term fluctuations. Bursts were identified as days where the tweet volume exceeded a specific statistical threshold:

$$Threshold = \text{Rolling Mean} + (2 \times \text{Rolling Std})$$

This approach successfully detected 122 significant bursts (outliers), representing periods of peak public reaction. The visualization of these results demonstrates that digital activity regarding

the Mati fire is non-linear, characterized by spikes triggered by specific events rather than a steady flow of discussion.

3.3 Author Activity Ranking (Task 3)

This analysis aims to identify active and influential authors by examining their temporal behavior during the Mati wildfire discussion. To reduce noise, authors with fewer than three relevant tweets were not included. Three metrics were computed for each author:

- Total tweet volume, measuring overall participation.
- Activity time span, defined as the number of days between the first and last tweet, capturing long-term involvement.
- Average time gap between tweets, measuring posting frequency and intensity.

These metrics were combined into a weighted activity score using logarithmic normalization to reduce the influence of extreme values and spam-like behavior. This approach allows fair comparison between authors with different activity scales.

$$Score = 0.5 \ln(\text{tweets}) + 0.3 \ln(\text{span} + 1) + 0.2 \left(\frac{1}{\ln(\text{gap})} \right)$$

The results reveal that high activity is not determined only by tweet volume, but also by sustained participation and frequent posting. Visualization through scatter plots highlights outliers and helps identify highly active or potentially automated accounts.

3.4 Time of Day Analysis (Task 4)

To understand daily rhythms, we extracted the hour of posting from the timestamps. We aggregated the tweet counts by hour across all dates to calculate the average activity per hour of the day. This highlights peak usage times, such as evening news cycles or active morning hours.

3.5 Engagement Metrics Aggregation (Task 5)

User engagement was analyzed to understand how the Twitter community interacted with content related to the Mati wildfire. Engagement metrics included likes, retweets, and replies, which were aggregated at both tweet and author level.

For each tweet, total engagement was calculated as the sum of likes, retweets, and replies. These values were then aggregated per author to measure overall influence and interaction.

The analysis shows that retweets dominate engagement, accounting for over 90% of all interactions. This reflects the nature of crisis-related events, where users primarily amplify information rather than create original content.

The scatter plot shows that engagement is unevenly distributed: a small number of authors receive very high engagement despite posting relatively few tweets, indicating that information diffusion is driven by influential users rather than by overall activity volume.

3.6 Time-Gap Analysis Between Tweets (Task 6)

We analyzed the time difference (Δt) between consecutive tweets for each author.

- **Rapid Posting:** Gaps < 1 minute were flagged to identify potential bot-like behavior or spamming.
- **Long Dormancy:** Gaps > 1440 minutes (1 day) were flagged to identify sporadic users.

The mean and standard deviation of these gaps were calculated for every author to assess consistency.

3.7 Weekly Activity Patterns (Task 7)

We mapped tweet volumes to a matrix of *Week of Year* vs. *Day of Week*. This data was visualized as a heatmap to show the intensity of discussion over specific calendar periods, allowing us to correlate social media activity with real-world events over the timeline.

3.8 Author Retweet Dependency (Task 8)

We classified tweets as "Retweets" if they started with the "RT" marker. We calculated a **Retweet Percentage** for each author:

$$\text{RT \%} = \left(\frac{\text{Retweets}}{\text{Total Tweets}} \right) \times 100$$

This distinguishes between original content creators (0% RT) and aggregators/amplifiers (100% RT).

3.9 Burst-Origin Analysis (Task 9)

To identify drivers of viral events, we isolated the timestamps of detected bursts (from Task 2). We analyzed the 6-hour window preceding each burst to identify "catalyst" authors. A **Burst Score** was assigned to authors based on:

$$\text{Score} = \frac{\text{Rank}(\text{Volume}) + \text{Rank}(\text{Time Lag})}{2}$$

This prioritizes authors who tweeted heavily *and* early relative to the burst.

3.10 Identification of Influential or Manipulative Authors (Task 10)

We extended the analysis to detect coordination and manipulation using four specific sub-tasks:

3.10.1 Top 1% Contributors (Sub-task 10.1)

Using the ranked dataframe from Task 9, we isolated the top 1% of authors based on their "average contribution" and "total burst participation". This establishes a baseline of the elite users who consistently drive volume during high-stress periods.

3.10.2 Catalyst Identification (Sub-task 10.2)

We refined the "catalyst" definition by examining the 6-hour pre-burst window. A user is flagged as a **Possible Catalyst** if their latest tweet in that window occurred within **1 hour** of the burst onset. This narrow threshold helps isolate the immediate triggers of viral spikes rather than general participants.

3.10.3 Coordination Patterns (Sub-task 10.3)

To detect artificial coordination (astroturfing), we analyzed "synchronized posting". We divided the timeline into 5-minute buckets and identified high-activity windows (top 90th percentile). An author's coordination was measured by:

1. **Sync Appearances:** How often the author appeared in these high-volume buckets.
2. **Coordination Score:** A composite metric combining volume, retweet dependency, and synchronization:

$$\text{Score} = 0.3(\text{Vol}_{\text{norm}}) + 0.4(\text{RT}\%) + 0.3(\text{Sync}_{\text{norm}})$$

This highlights users who do not just post often, but post *at the exact same time* as others, a strong indicator of coordinated campaigns.

3.10.4 Network Roles & Burst Amplification (Sub-task 10.4)

We classified users based on **Burst Amplification**, defined as the ratio of their activity share during bursts vs. the burst's duration share in the dataset:

$$\text{Amplification} = \frac{\text{User Burst Share}}{\text{Burst Time Share}}$$

Users were categorized into three behavioral roles based on this score:

- **Normal Activity** (≤ 1.5): Users whose activity is consistent regardless of bursts.
- **Mixed Activity** ($1.5 < x < 3.5$): Users who show moderate increase during crises.
- **Suspicious Activity** (≥ 3.5): Users who are disproportionately active *only* during bursts, suggesting "sleepers" accounts or bots activated for crisis amplification.

4 Results and Analysis

4.1 Temporal Analysis and Bursts

The daily volume of tweets shows distinct spikes corresponding to the event. The rolling average highlights the baseline activity, while the burst detection algorithm successfully identified days with statistically significant deviations.

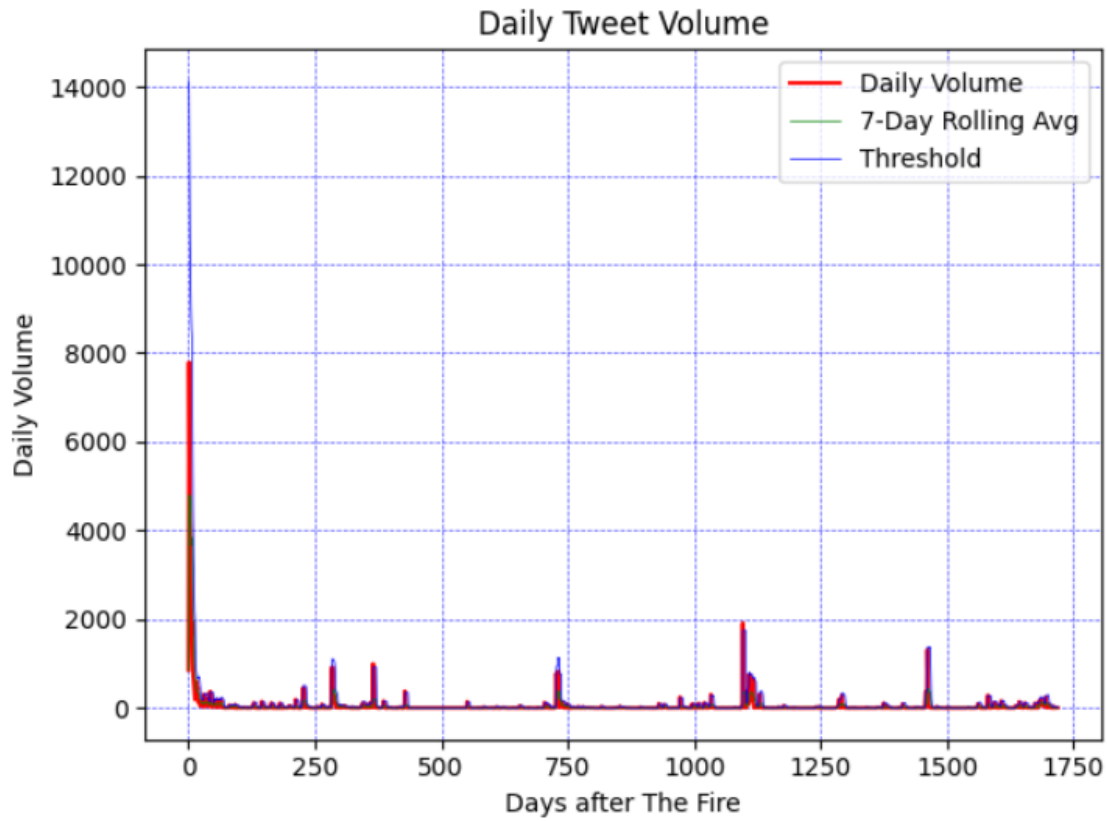


Figure 1: Daily tweet volume with 7-day rolling average and threshold.

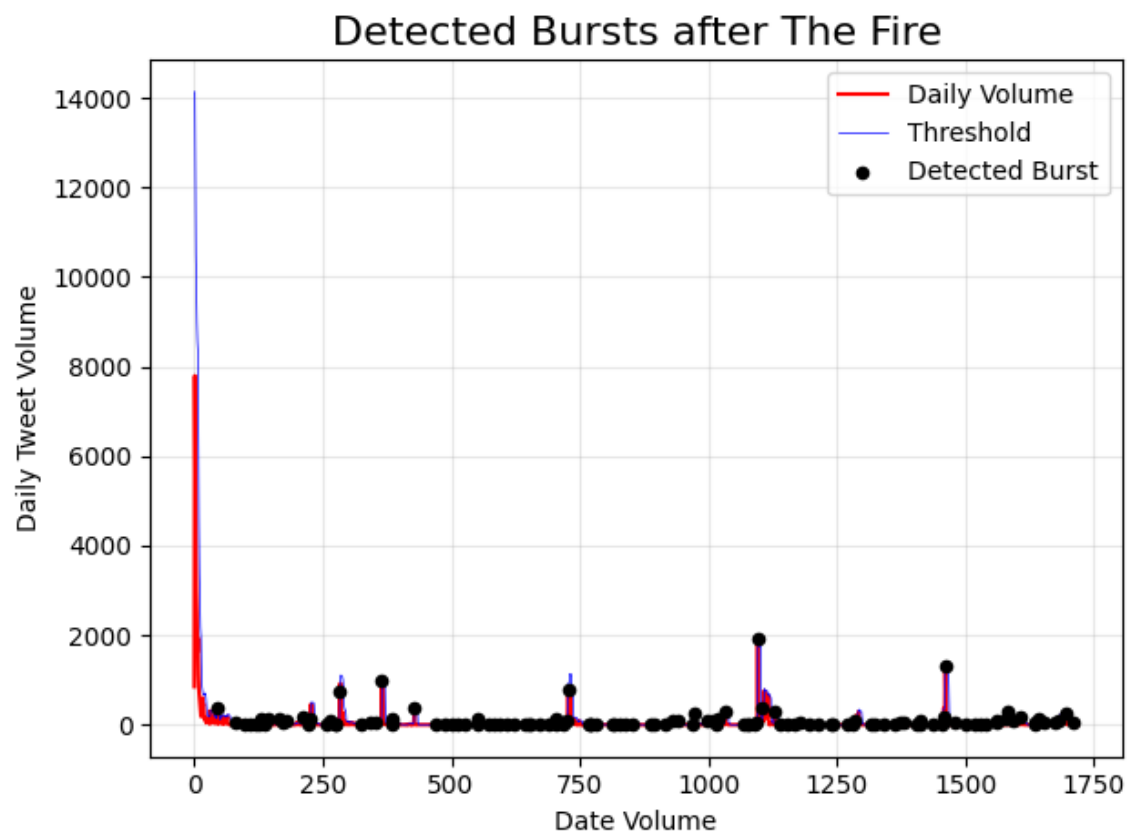


Figure 2: Detected bursts where volume exceeded the threshold.

4.2 Author Activity and Trends

The weighted activity score highlights users who are consistently active over long periods rather than just high-volume spammers. The trend line indicates the average relationship between volume and the composite score.

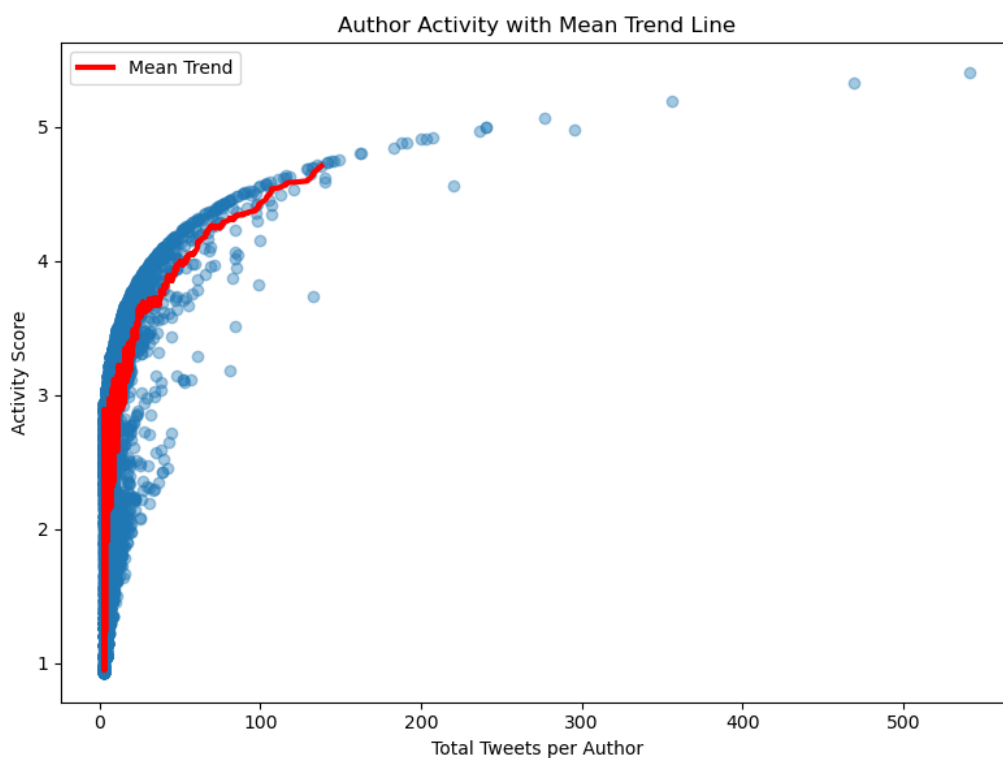


Figure 3: Author Activity Score vs. Total Tweets with mean trend line.

4.3 Time and Weekly Patterns

Activity on Twitter regarding the fire follows a clear daily pattern. Additionally, the weekly heatmap reveals the specific weeks where the conversation was most intense.

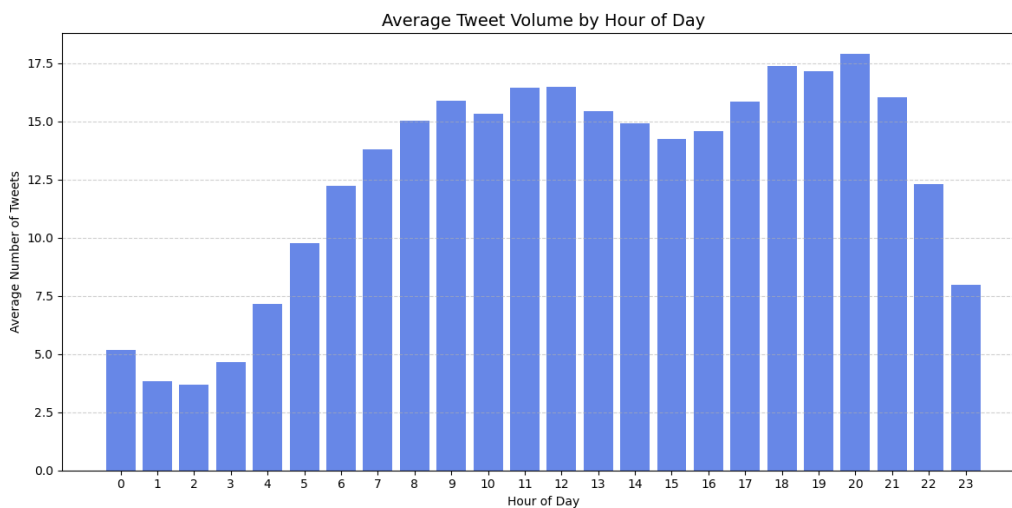


Figure 4: Average tweet volume by hour of day.

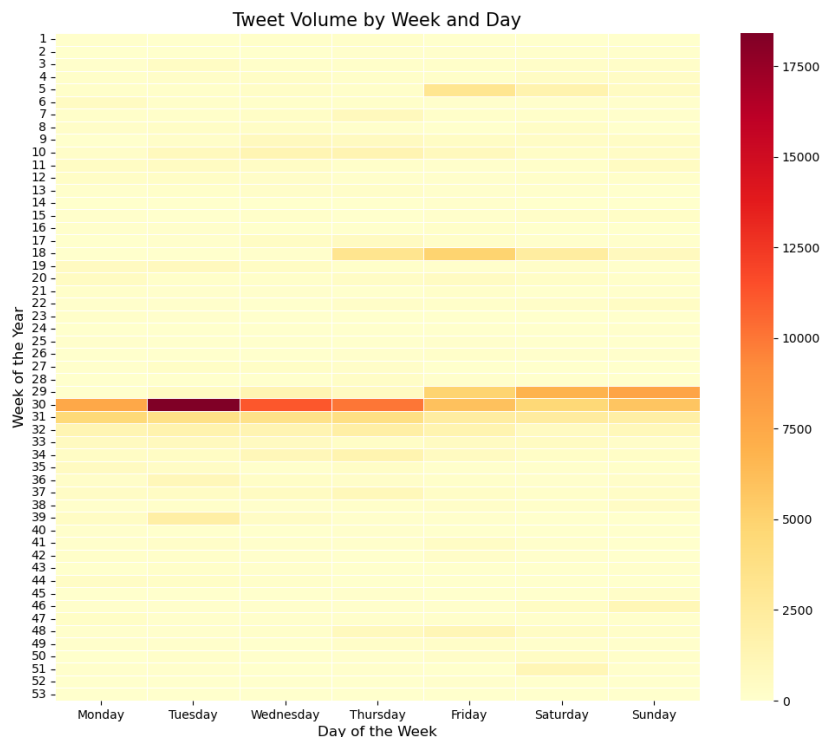


Figure 5: Heatmap of tweet volume by Week of Year and Day of Week.

1. In the heatmap we can see a clear surge in tweets, the most intense of which during the 30th week of the year (23rd of July-30th of July) and specifically Tuesday, one day after the incident, and the days around it. This can be explained because most casualties were reported during late night hours, so its logical that the surge occurred the day after. We can also see a clear yearly pattern around this week. Around the darkest point of the heatmap there are major events regarding the tragedy such as yearly memorials, relatives posts, political confrontations etc.
2. Also we can see a clear surge in the 18th of the week, which is the date of the release of a major documentary about the incident in 2019, which clearly sparks conversation in the social network. Also during the same days, the trial for the Mati incident began. 28 April which is a surge, marks the beginning of the actual trial for the Mati incident.
3. In addition, we can see a surge during the last week of January. This marks the beginning of the trial process for the Mandra floods, connecting the incident with the mati tragedy.
4. One important aspect is the trial process targeted the same ministers that were responsible for the mati tragedy.

4.4 Engagement and Content Distribution

The engagement analysis shows how the community interacted with the content. The distribution of likes, retweets, and replies provides insight into the viral nature of the dataset.

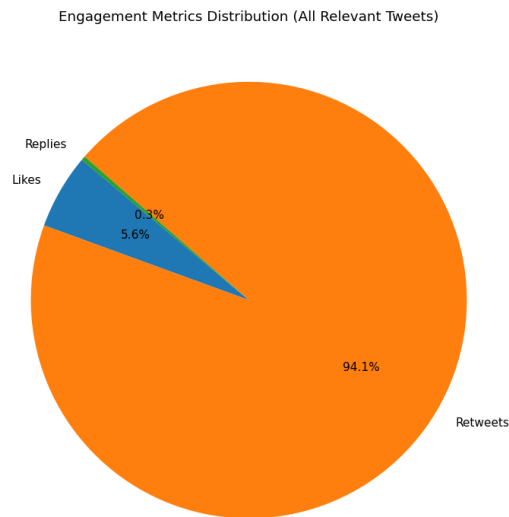


Figure 6: Distribution of Engagement Metrics.



Figure 7: Total Engagement vs. Total Tweets.

4.5 Retweet Dependency

We analyzed how much the conversation relied on retweeting versus original content creation. The Global Percentage of Retweets (Figure 8) reveals that **76.7%** of all relevant tweets in the dataset are retweets (159,319), while only 23.3% are original posts. This indicates that the information landscape regarding the Mati wildfire is dominated by amplification and dissemination rather than the generation of new information.

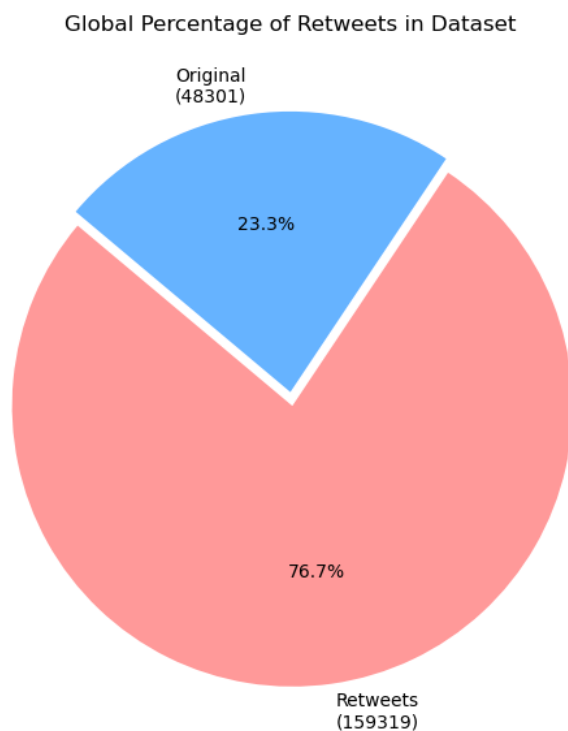


Figure 8: Global percentage of Retweets vs Original Tweets.

The *Distribution of Author Retweet Dependency* (Figure 9) further clarifies this behavior. The histogram displays a bimodal distribution: a significant number of authors have a Retweet Percentage of near 0% (pure content creators), while an even larger spike exists at **100%** (over 17,500 users). This suggests that a vast portion of the user base functions exclusively as "amplifiers," possibly indicating casual users or, in some cases, automated accounts dedicated to boosting visibility.

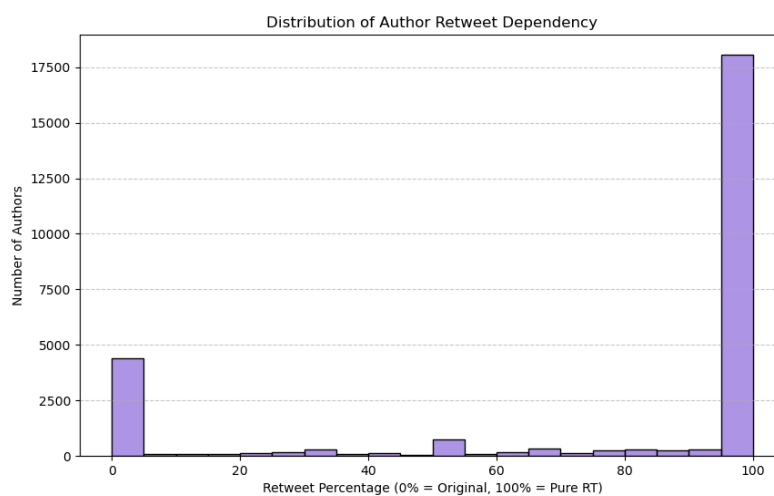


Figure 9: Distribution of Author Retweet Dependency.

Finally, the scatter plot of *Activity Volume vs. Retweet Dependency* (Figure 10) demonstrates a correlation between high volume and high retweet percentage. The users with the highest activity counts (far right on the log-scale x-axis) tend to cluster towards the upper ranges of retweet dependency.

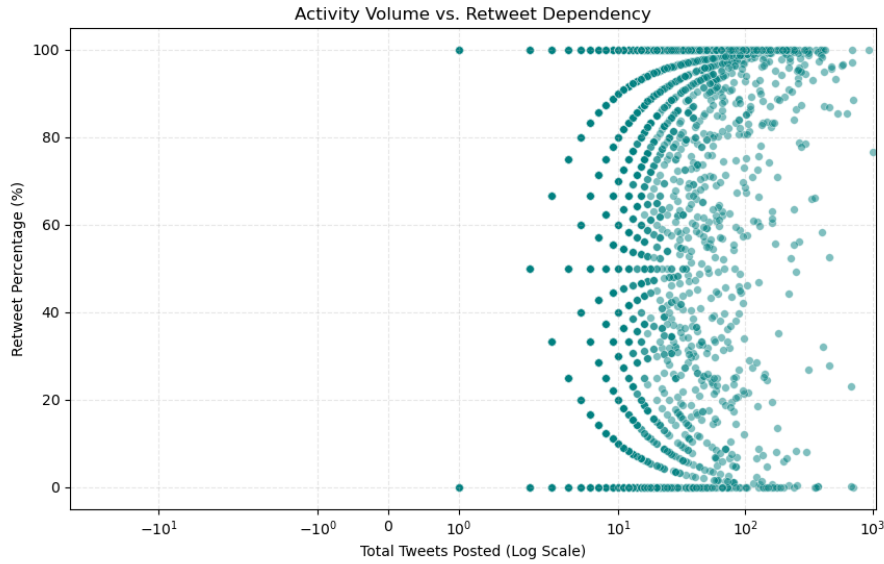


Figure 10: Activity Volume vs. Retweet Percentage.

4.6 Advanced Analysis: Coordination and Network Roles

In Task 10, we investigated signs of coordinated behavior and user roles relative to crisis bursts. The *Hourly Activity of Top Coordinated Authors* heatmap (Figure 11) visualizes the specific hours where high-score authors were active. Dark red cells indicate high intensity, but more importantly, distinct vertical alignments (e.g., around hours 18-20) suggest synchronization. Multiple top authors are active simultaneously during specific windows, which is a hallmark of coordinated campaigns or highly synchronized reactions to external media events.

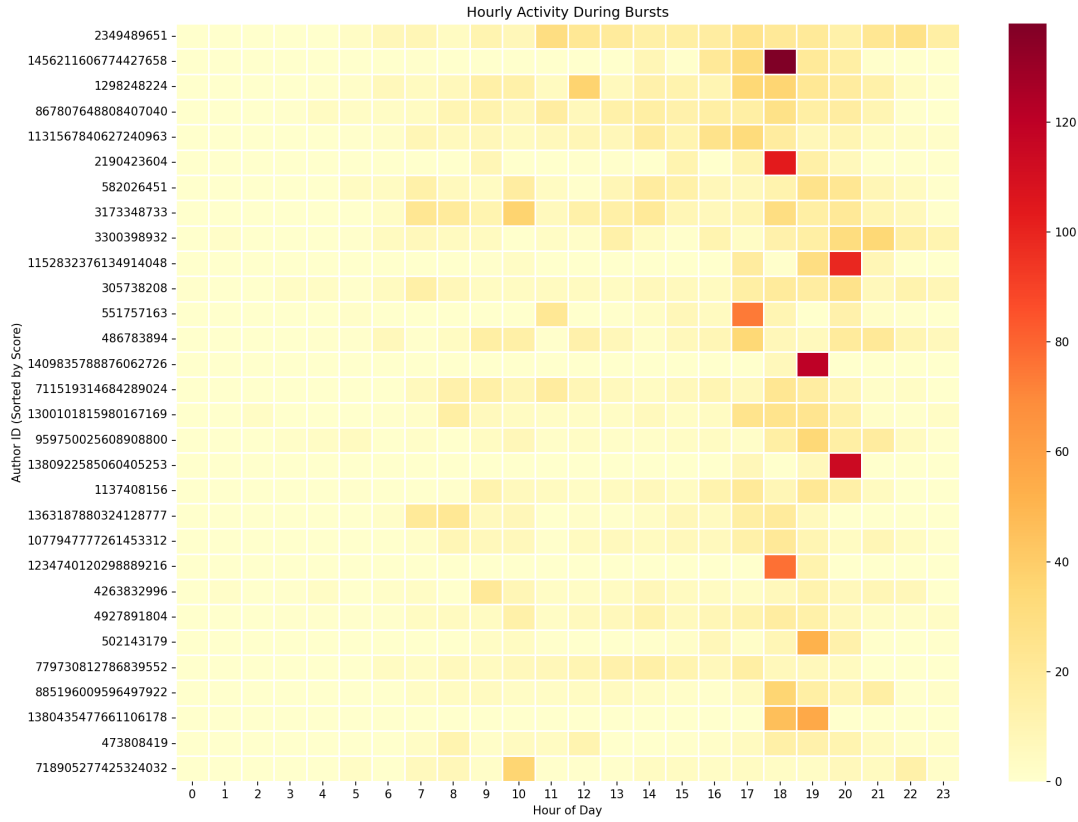


Figure 11: Hourly activity of top coordinated authors during bursts.

We further classified users by comparing their activity during bursts versus normal periods. The *Top Authors by Burst Volume* bar chart (Figure 12) highlights users like *harhalas* and *anatheniantoldn*, who exhibit massive activity during bursts (Red bars) compared to relatively low activity during normal periods (Blue bars).

This distinction is formalized in the *Network Roles* scatter plot (Figure 13). Here, users are plotted based on their Normal Volume vs. Burst Volume.

- **Normal Activity (Blue):** Users clustered below or near the diagonal dashed line ($y = x$) behave consistently regardless of the news cycle.
- **Suspicious Activity (Red):** Users clustered significantly above the diagonal line are disproportionately active during bursts. These "sleeper" or "agitator" accounts drive the viral spikes observed in the dataset.

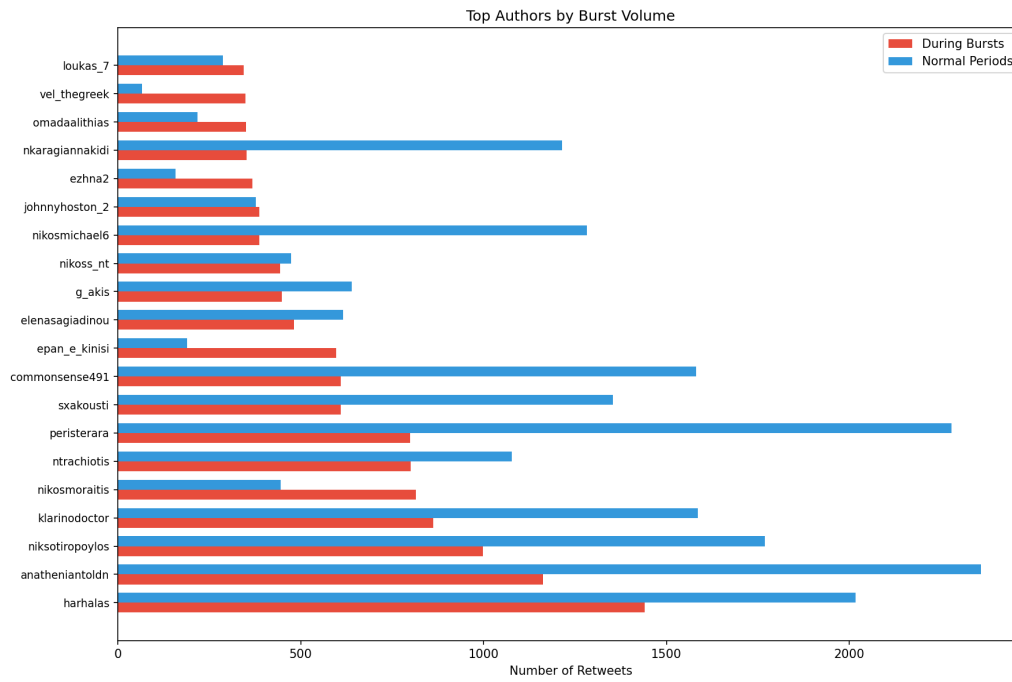


Figure 12: Top Authors by Burst Volume: Comparison of Burst vs Normal activity.

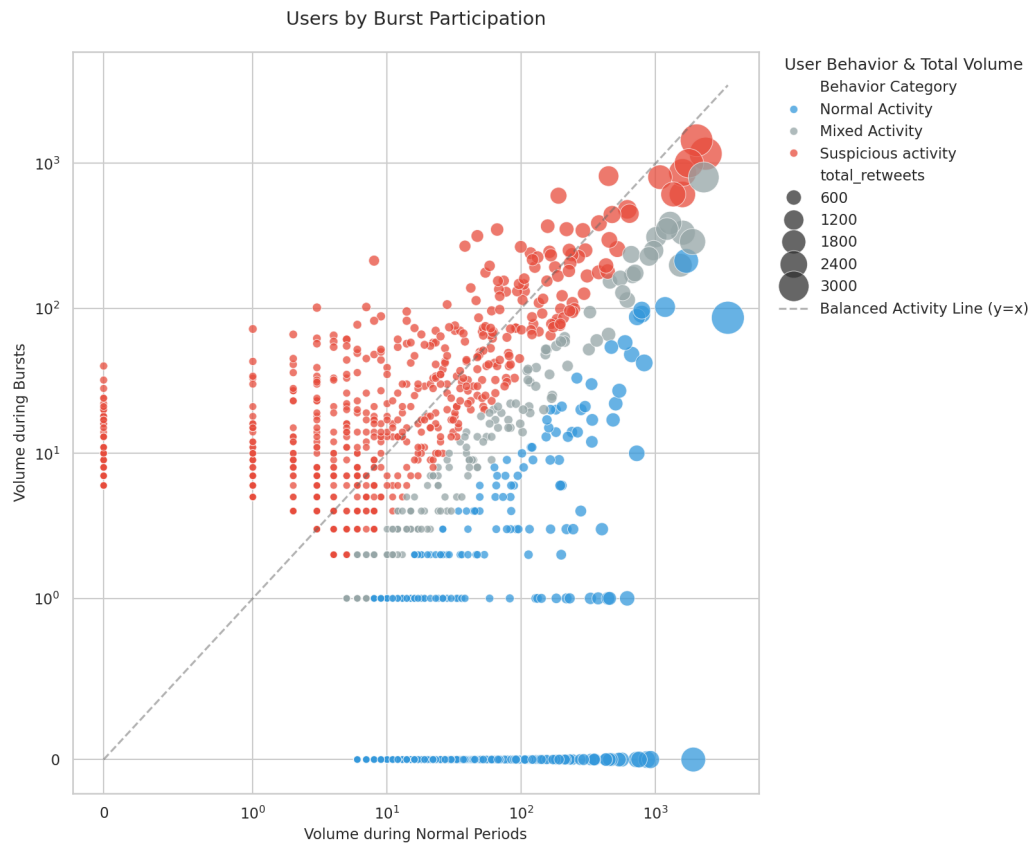


Figure 13: Network Roles: Classifying users by burst amplification (Suspicious vs Normal).

5 Conclusion

The analysis provided a comprehensive view of the Twitter landscape during the Mati wildfire. By combining temporal analysis, engagement metrics, and advanced coordination detection, we

were able to distinguish between organic public outcry and potential coordinated amplification. The identification of "catalyst" authors and the separation of original content creators from retweeters provides valuable data for understanding social media dynamics in crisis situations.